UNIVERSITY OF LONDON

BSc EXAMINATION 2018

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS3003: IT Project Management

Marking Scheme

Time Allowed: **TWO** hours

Answer THREE questions
No credit will be given for attempting any further questions

Calculators are NOT permitted

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General Note

The exam questions are structured in such a way that approximately 50% of the marks come through bookwork, while for the remainder students need to apply the knowledge, analytical and judgment skills gained in the course to make decisions or solve new problems.

Summary of modules' learning outcomes

LO1 - Understand the fundamental concepts, terminology and cost elements of IT Projects.

LO2 - Apply basic Project Planning techniques.

LO3 - Explain the procedures needed to monitor, control and report upon an IT Project.

LO4 - Demonstrate an understanding of the steps needed to build and maintain effective development teams.

LO5 - Discuss and where appropriate apply the principles of Project Risk Management.

LO6 - Explain the ways in which appropriate quality attributes of the products of an IT Project can be assessed and assured.

Mapping of Exam Questions to Learning Outcomes

Question Number	Learning Outcome
1a	*LO3*
1b	*LO1*
1c	*LO3*
1d	*LO1*
1e	*LO2*
2a	*LO2, LO3*
2b	*LO2*
2c	*LO6*
3a	*LO1*
3b	*LO1*
3c	*LO1*
3d	*LO4*
4a	*LO4*
4b	*LO5*
4c	*LO1, LO3*
4d	*LO3*

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1.

(a) In the context of configuration management, indicate whether each of the following statements is **true** or **false**. A justification is **not** required.

(i) Change management is an activity of configuration management;
(ii) Effective configuration management requires specialized tools;
(iii) System building is the process of assembling components into an executable form;
(iv) Software documentation is a valid configuration item;
(v) It is an appropriate practice for different projects to use their own, project-specific, source code version management systems.

[10 marks]

Marking Scheme *(partially bookwork, students should demonstrate appropriate understanding of bookwork)*

2 marks per correct answer

(b) In the context of project budgeting, explain the difference between contingency and management reserves. [6 marks]

Marking Scheme *(partially bookwork, students should demonstrate appropriate understanding of bookwork)*

6 marks for correct, comprehensive, well-explained answer. Partial marks for answer that makes sense but not sufficiently justified/explained.

(c) You have been assigned as the Project Manager of a project that is significantly delayed. In order to cope with the delay, the project sponsor has asked the project teams to work paid overtime. Suggest three additional actions that may be performed to bring the project back on schedule. [6 marks]

Marking Scheme (*mostly bookwork*).

2 marks for each correctly identified action.

(d) One of the advantages of package-based software is "*the opportunity to obtain business know-how and adopt best-practices which are typically embedded in a package*". Explain in your own words how you understand the reasoning for the above statement. Also, provide one example that illustrates its validity in the context of the functionality provided by a software application of your choice.

Note: An extensive analysis is not required but your answer should be sufficiently self-contained to make sense. [7 marks]

Marking Scheme *(non-bookwork)*

5 marks for appropriate justification. 2 marks for example that makes sense.

(e) You have been assigned as the Project Manager to a new project and you have been given a preliminary project scope statement. Which of the four tasks below is the one you will perform first?

- Analyse project risk.
- Prepare an initial project plan.
- Perform stakeholders' analysis and confirm that all stakeholders have contributed to the project scope.
- Initiate the requirements analysis phase.

Briefly **justify** your answer. [5 marks]

Marking Scheme (*non-bookwork*)

3 marks for correct answer, 2 marks for justification.

2.

(a) One of the Project Manager responsibilities is to manage project communications.

(i) Provide the definition of Project Communications Management; [3 marks]

(ii) Refer to four elements that a Project Communications Plan should include; [4 marks]

(iii) Describe any two actions a Project Manager may take in order to develop an appropriate Project Communications Plan. [4 marks]

Marking Scheme *((i) and (ii) bookwork, (iii) non-bookwork)*

(i) Full marks for correct and comprehensive answer; (ii) 1 mark for each correctly identified element of a project communications plan; (iii) 2 marks for each action that makes sense

(b) The activities of a project and respective durations and dependencies are shown in the following table.

Activity	Duration (weeks)	Predecessors
A	3	-
B	10	A
C	8	A
D	11	A
E	1	B,C
F	2	C
G	2	E,F
H	1	G,D

(i) Draw a network diagram for this project including all floats; [7 marks]

(ii) Mark the project's critical path on the diagram; [2 marks]

(iii) Draw the project's Gantt chart; [4 marks]

(iv) Identify two advantages of using the project Gantt chart, in comparison to the network diagram, in what concerns the information provided on the planning and progress of a project. [4 marks]

Marking Scheme (*non-bookwork*)

i. 1 mark for each correct node/sequence/float in the diagram, excluding node A. Deduction of 2 marks for each wrong node/float

ii. Full marks for correct critical path

iii. 0.5 marks for each correct representation of activity in the bar chart

iv. 2 marks for each correctly identified advantage

(c) In the context of project quality, explain the main differences between quality assurance and project assurance. [6 marks]

Marking Scheme (*bookwork*)

4 marks for adequate description of the difference in scope. 2 marks for the fact that quality assurance is independent of the project and that project assurance is independent of the project manager.

3.

(a) A project consists of the following three activities: Activity A with a planned value (PV) of £500, Activity B with a planned value of £1000 and Activity C with a planned value of £100. During the execution of the project, the following figures have been assessed: Activity A is complete and had a cost of £500. Activity B is 50% complete and has an actual cost of £700. Activity C is 75% complete and has an actual cost of £90.

(i) What is the total Earned Value (EV) of the project? [6 marks]

(ii) What is the Cost Performance Index (CPI) of the project? [4 marks]

Marking Scheme (*non-bookwork*)

(i) Full marks for correct calculation of EV. Partial marks depending on correct steps. (ii) full marks for correct calculation of CPI. 2 marks for providing CPI formula only.

(b) *"In some IT projects the system architecture may have to be designed before the requirements specification is complete"*. Do you agree with this statement? Briefly **justify** your answer. [7 marks]

Marking Scheme *(non-bookwork)*

3 marks for "agree". 4 marks for appropriate justification. Two reasons will suffice for full marks.

(c) In comparison to projects in other domains, one of the distinguishing characteristics of IT Projects is that the end-product is intangible. Discuss three additional distinguishing characteristics of an IT Project. [9 marks]

Marking Scheme *(bookwork)*

1 mark for each correctly identified distinguishing characteristic of an IT Project. 2 marks for appropriate explanation/discussion per characteristic.

(d) "Poor leadership" is one of the main factors that project teams fail or do not perform adequately in an IT Project. Indicate four additional, widely-accepted, factors that result in ineffective project teams. [8 marks]

Marking Scheme *(bookwork)*

2 marks for each correctly identified factor.

4.

(a) Two software developers are involved in a conflict that is threatening to disrupt an IT project. The project manager calls the two developers into her office and reminds them that they are both *"on the same side/team in working towards the project's success"*. Which method of conflict resolution is the project manager applying in this case?. Briefly **justify** your answer.

[6 marks]

Marking Scheme *(non-bookwork)*

Full marks for correct answer (Mediate or Defuse) – 2 marks for an incorrect answer which was sufficiently justified.

(b) Draw a diagram depicting the four generic activities of a risk management process. Outline the tasks performed in each activity and the respective outcomes. [12 marks]

Marking Scheme *(bookwork)*

4 marks for complete diagram (1 mark per correct activity identified). 2 marks per activity for correctly identifying outcomes and providing an outline of tasks performed.

(c) One of the responsibilities of an IT Project Manager is to ensure that appropriate documentation is handed over at the rollout/go-live phase of a project, for instance, the go-live "sign-off" document. Indicate three additional documents that should be handed over at the rollout/go-live phase of an IT Project, explaining the purpose of each one. [9 marks]

Marking Scheme *(non-bookwork)*

3 marks for each document identified and for explaining its purpose. 2 marks for document without an explanation of purpose.

(d) You are the project manager in an ongoing IT project in your organization. Your boss has asked you to prepare a report on the project progress which will be communicated to senior management. Indicate seven different types of information that such report is expected to include. [7 marks]

Marking Scheme *(partially bookwork)*

1 marks per type of information that a project progress report is expected to include